

Problem 8.4.3

$$H_p = 0.95, H_s = 0.15, \omega_s = 500 \text{ rad/s}, \omega_p = 1000 \text{ rad/s}$$

$$\omega_s = 500/500 = 1 \text{ rad/s}, \omega_p = 1000/500 = 2 \text{ rad/s}$$

$$\omega_{p0} = 1/\omega_p = 0.5 \text{ rad/s}$$

$$\varepsilon_{ps} = \sqrt{\frac{H_s^2}{1 - H_s^2}} = 0.152$$

$$aa = \sqrt{\frac{1}{\varepsilon_{ps}^2} * \frac{1}{1 - H_p^2}} = 21.1 \quad bb = \frac{1}{\cosh^{-1}(\frac{1}{\omega_{p0}})} = 0.759$$

$$n = \cosh^{-1}(aa) * bb = 2.84 \cong 3$$

$$\alpha = \frac{1}{\varepsilon_{ps}} + \sqrt{1 + \frac{1}{\varepsilon_{ps}^2}} = 13.3$$

$$a = 0.5 * \left(\alpha^{\frac{1}{n}} - \alpha^{-\frac{1}{n}} \right) = 1.04 \quad b = 0.5 * \left(\alpha^{\frac{1}{n}} + \alpha^{-\frac{1}{n}} \right) = 1.44$$

$$p_0 = a = 1.04 \quad d_0 = s + \frac{1}{a} = s + \frac{1}{1.04}$$

$$\theta = 60^\circ * \frac{\pi}{180} = 1.047 \text{ rad} \quad p_1 = a \cos \theta + j b \sin \theta = 0.520 + j1.25$$

$$q_1 = \frac{1}{p_1} = 0.004 - j(2 * 10^{-5})$$

$$d_1 = s^2 + (2 * 0.004)s + (0.004^2 + (2 * 10^{-5})^2) = s^2 + 0.008s + (1.6 * 10^{-5})$$

$$\omega_1 = \frac{1}{\cos(\frac{\pi}{2 * 3})} = 1.15$$

$$num = s^2 + \omega^2 \quad den = d_0 d_1$$

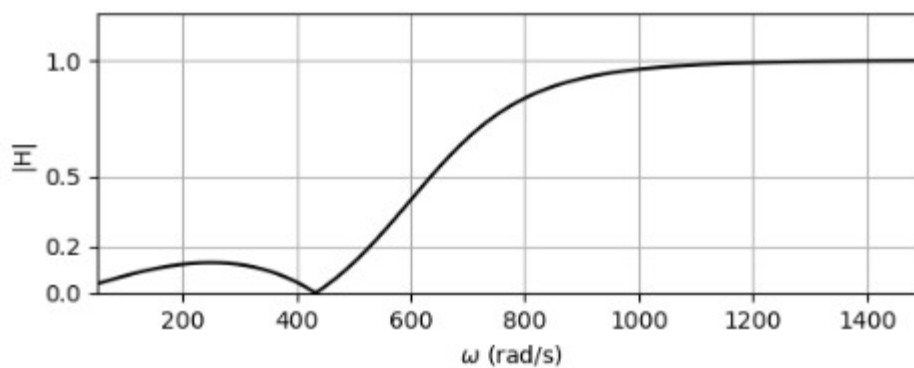
$$H(s)_{LPF} = \frac{K * (s^2 + 1.32)}{(s + 0.962)(s^2 + 0.008s + (1.6 * 10^{-5}))}$$

$$K = \frac{3.2 * 10^{-8}}{1.32} = 2.4 * 10^{-8}$$

$$H(s)_{HPF} = \frac{(3.2 * 10^{-8})s^3 + (2.4 * 10^{-8})s}{(3.2 * 10^{-8})s^3 + (3.2 * 10^{-5})s^2 + (10 * 10^{-3})s + 1}$$

To scale back up: $a = 500$

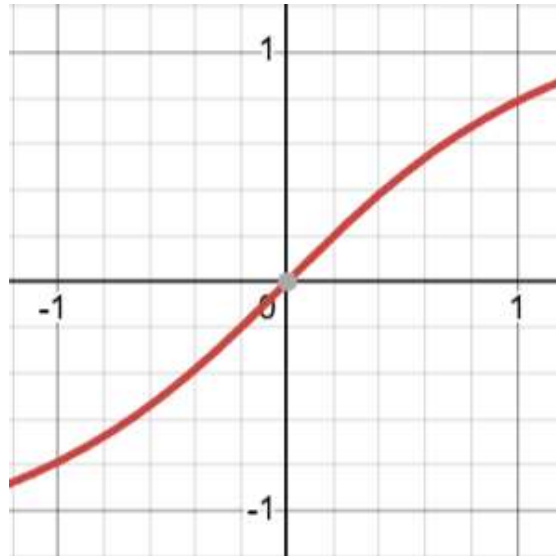
$$\begin{aligned}
 H(s)_{final} &= \frac{(3.2 * 10^{-8})s^3 + a^2(2.4 * 10^{-8})s}{(3.2 * 10^{-8})s^3 + a(3.2 * 10^{-5})s^2 + a^2(10 * 10^{-3})s + a^3} \\
 &= \frac{(3.2 * 10^{-8})s^3 + 0.006s}{(3.2 * 10^{-8})s^3 + 0.016s^2 + 2500s + (1.25 * 10^8)}
 \end{aligned}$$



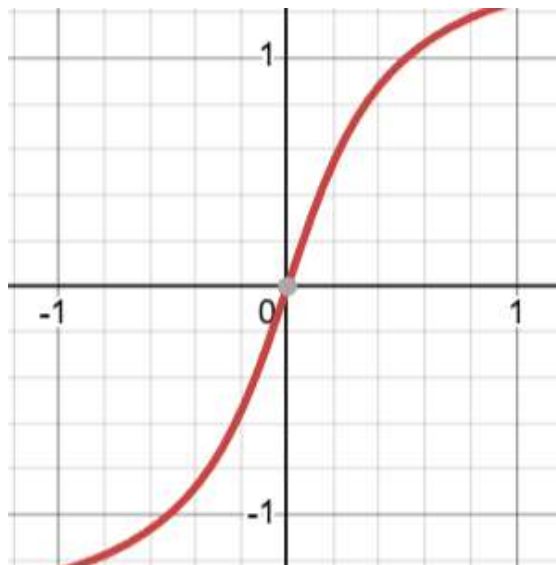
Problem 8.6.1

Since $B_0(s) = 1$, $\varphi = \tan^{-1}(0) = 0$

Since $B_1(s) = s + 1 = j\omega + 1$, $\varphi = \tan^{-1}(\omega)$,
which is linear when ω is small as shown below.



Since $B_2(s) = s^2 + 3s + 3 = -\omega^2 + 3j\omega + 3$, $\varphi = \tan^{-1}(3\omega)$,
which is linear when ω is small as shown below.



As the polynomials continue, the relationship between ω and φ is just $\varphi = \tan^{-1}(k\omega)$, with k increasing each time. This inevitably still leaves a linear relationship between ω and φ as long as ω is small.